

Generation Interconnection System Impact Study Report

For Queue Project Q23-2357

South Bluefield 138 kV

March 03, 2025

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q23-2357. The Interconnection Customer proposes a 37.2 MW battery energy storage facility interconnecting at the South Bluefield 138 kV substation in the Virginia area. The request was received on November 13, 2023. The study evaluated system intact and single contingency conditions using the regional planning case, including 494 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 6 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$330,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
International Paper Valliant Ok / W Winchester 138 kV line	International Paper Valliant Ok / Kearneyville 345 kV	114.0	636	7.0
Alvin / Nuclear Fuels 138 kV	Purdy / Page 138 kV	113.9	310	8.8
St. John / Rio Dell 138 kV L	International Paper Valliant Ok / Kearneyville 345 kV	112.5	593	6.6
Jones Chapel / Bluebonnet 230 kV line	Robison Park / Russell Fork 230 kV	106.5	2,635	6.1
Boca Raton / Pueblo Xtra 500 kV	Bluebonnet / Jones Chapel 230 kV	105.5	3,392	9.0
Concord / Bluebonnet 230 kV	Pueblo Tra / Boca Raton 500 kV	104.0	1,853	6.5

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
W Winchester / International Paper Valliant Ok 138 kV	Rebuild line with higher-capacity conductor	\$1,270,000	2% (\$19,000)
Nuclear Fuels / Alvin 138 kV Line	Rebuild line with higher-capacity conductor	\$5,820,000	1% (\$80,000)
St. John / Rio Dell 138 kV L	Rebuild line with higher-capacity conductor	\$5,102,000	2% (\$118,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Bluebonnet / Jones Chapel 230 kV	Reconductor line	\$8,789,000	0% (\$19,000)
Pueblo Xtra / Boca Raton 500 kV L	Reconductor line	\$36,072,000	0% (\$77,000)
Bluebonnet Substation / Concord 230 kV	Reconductor line	\$7,920,000	0% (\$17,000)
	Total allocated to project		\$330,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.